

PROBABILITY DISTRIBUTIONS

● INTRODUCTION : Why Probability Distributions Matter.

- Heart of Data Analytics.
- They provide a mathematical framework to describe uncertainty, model real world phenomena and make informed predictions.

Whether you are analyzing customer purchases, forecasting rainfall, or building a machine learning model, understanding probability distribution is essential for interpreting data

- recognizing patterns &
- making sound decisions.

* WHAT IS PROBABILITY DISTRIBUTION :-

- A PROBABILITY DISTRIBUTION is a mathematical function that describes how likely [possibility to happen] different outcomes are for a random event or variable.
- In other words, it tells us how probabilities are spread across all possible values that a variable can take.

⇒ SIMPLE ANALOGY :-

RANDOM
PROCESS

Imagine you have a bag of coloured marbles — RED
BLUE
GREEN
If you pick one marble at random, the probability distribution tells you the chance of picking each colour.

If there are more red marbles, the PROBABILITY of picking RED is HIGHER.

RANDOM
VARIABLE

* KEY POINTS :

- A probability distribution assigns a probability (a number between 0 $\leftrightarrow$ 1) to each possible outcome.
- The sum of all probabilities is always 1 (or 100%)
- DISTRIBUTIONS can describe simple events (like rolling a die) or complex phenomena (like daily rainfall/customer spendings).

* WHY IS THIS IMPORTANT :

PROBABILITY DISTRIBUTIONS help us answer questions like :-

- What is the most likely outcome?
- How rare or common are certain values?
- How spread out or clustered is the data?
- What are the chances of something unusual?

* RANDOM VARIABLE

- A random variable is a variable whose value is determined by the outcome of a random process.
- It can be :-
 - DISCRETE :- Takes specific, countable values (e.g., NO. OF HEADS in 10 coin TOSSES) whole number
 - CONTINUOUS :- Can take any value within a range. not whole no
(E.g., Height of a person - 165.85 cm ↙
Amount spent in a store \$3865.50)

* THINK RANDOM VARIABLE as a BOX that holds the result of a random experiment - like the number we get while rolling a die.